Robed Beauvile-Owen

Vandergrift, PA | 850-354-0565 | robedbeauvile@gmail.com | linkedin.com/in/robedbeauvile | github.com/beauvilerobed

EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy delivering site-selection analyses for retail and service clients, including a $1.5M expansion opportunity.
- Estimated the incremental revenue of new locations with Double ML and causal forests (econml, XGBoost nuisance models) on a 12-year county panel, validated by DiD and a synthetic-data recovery test.
- Engineered Python ETL pipelines automating geospatial data ingestion and reporting, cutting market analysis turnaround by 91% (weeks to days).
- Built a Monte Carlo risk framework modeling correlated demand, cost, and competition inputs, letting clients stress-test expansion decisions.
- Designed a proprietary site-scoring framework integrating U.S. Census demographic data, traffic patterns, competitive density, consumer behavior, and geospatial analytics to estimate revenue potential and location risk.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Led the healthcare marketing analytics and data science roadmap in partnership with strategy, engineering, and behavioral science leadership, establishing OKRs and strategic analytics initiatives.
- Applied predictive modeling, customer segmentation, and uplift modeling to identify high-value Medicare/Medicaid eligible cohorts, reducing ad spend by $10K per campaign across a pipeline of millions of healthcare profiles.
- Designed, deployed, and standardized cross-team experimentation and causal inference frameworks for direct-mail campaign optimization, applying CUPED to reduce metric variance by 20%, enabling smaller samples and faster test reads.
- Owned a monthly model monitoring report tracking drift and performance for up to 10 production models, including media mix, response, eligibility, and psychology-match models.
- Re-architected an end-to-end distributed churn prediction platform using Python, SQL, Snowflake, Spark, and Databricks, reducing runtime by 48% and memory consumption by 23% via pipeline code optimization.

Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- Defined quantitative success metrics, performance evaluation frameworks, and drift monitoring for production search and regression models using offline and online experimentation.
- Developed a regression-based shipping price API that accurately breaks down bulk order history to estimate individual item shipping costs; scaled the system to handle 10,000+ daily requests while maintaining strict performance standards.
- Reduced search customer dissatisfaction by 12% with product and engineering teams via custom Elasticsearch indexing, tokenizers, and TF-IDF weighting.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Developed and evaluated XGBoost and other classification models using millions of healthcare consumer records and hundreds of appended external attributes, driving a 3% lift in response rates.
- Streamlined reporting and visualization analytics pipelines via Python, SQL, Snowflake, and Power BI; removed 60 minutes of manual daily overhead while strengthening data scalability.
- Improved geospatial targeting accuracy by 20% through GIS analysis with GeoPandas and spatial modeling, sharpening opportunity identification.

Graduate Researcher & Teaching Assistant — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Conducted original doctoral research spanning mathematical optimization, functional analysis, and operator theory, producing 3 substantial results addressing complex theoretical problems to complete doctoral dissertation requirements.
- Taught undergraduate mathematics and statistics courses (75-200 students) and coordinated summer programs for 100+ students (95% pass rate).

TECHNICAL PROJECTS

Quantitative Economics Platform github.com/beauvilerobed/quantitative-economics-with-python

Technologies: Jupyter, NumPy, SciPy

- Built an open-source research platform modeling stochastic dynamics with Markov chains, survival analysis, and Kalman filtering, the state-transition methods behind customer retention and churn forecasting.

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

Technologies: Python, NumPy, Survival Analysis, Monte Carlo Simulation

- Simulated 1M customer lifetimes under constant and tenure-dependent churn hazards; showed the LTV = ARPU / churn shortcut fails when churn varies with tenure and estimated customer lifetime value from empirical survival curves.

CaLaun — *AI-Powered Calculus Tutor Platform* github.com/calaun-learning/calaun

Technologies: Python, LLM Integration, Django, PostgreSQL

- Designed an open-source calculus tutoring platform that combines symbolic mathematics with large language models to generate accurate, step-by-step calculus solutions, serving 400+ weekly users with 90+ automated tests to secure deployment reliability.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90 | **Undergraduate Thesis:** Homotopy Type Theory, Univalent Foundations, and Binary Trees

TECHNICAL SKILLS

Programming: Python, SQL, R, C#, Bash, PowerShell

Statistics & Experimentation: Causal Inference, Double ML, Causal Forests, A/B Testing, CUPED, Power Analysis, Hypothesis Testing, Difference-in-Differences, Synthetic Controls, CATE

Machine Learning: XGBoost, Scikit-learn, Regression, Classification, Clustering, Uplift Modeling, Media Mix Modeling, Monte Carlo, Model Monitoring, LLMs

Data Engineering & Cloud: Spark, Databricks, Snowflake, AWS, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL, Feature Engineering

Visualization & Leadership: Power BI, Tableau, Excel, Matplotlib, Plotly, Stakeholder Management, Cross-Functional Teams, OKRs/KPIs